

# Deep Learning Practical Oral Questions and Answers

## 1. Linear Regression using Deep Neural Network (Boston Housing Dataset)

**Q1. What is Linear Regression?**

**Answer:** A supervised learning algorithm used for predicting continuous values.

**Q2. What is Deep Neural Network?**

**Answer:** A neural network having multiple hidden layers.

**Q3. What is Boston Housing Dataset?**

**Answer:** A dataset used for predicting house prices based on different features.

**Q4. What is supervised learning?**

**Answer:** Learning using labeled input and output data.

**Q5. What is regression?**

**Answer:** Predicting continuous numerical values.

**Q6. What is a feature?**

**Answer:** An input variable used for prediction.

**Q7. What is target variable in Boston dataset?**

**Answer:** House price.

**Q8. What is an epoch?**

**Answer:** One complete iteration over the training dataset.

**Q9. What is batch size?**

**Answer:** Number of samples processed before updating weights.

**Q10. What is activation function?**

**Answer:** Function applied to neuron output.

**Q11. What activation is used in regression output layer?**

**Answer:** Linear activation.

**Q12. What is loss function for regression?**

**Answer:** Mean Squared Error (MSE).

**Q13. What is optimizer?**

**Answer:** Algorithm used to update weights.

**Q14. What is Adam optimizer?**

**Answer:** Adaptive optimization algorithm used in deep learning.

**Q15. What is overfitting?**

**Answer:** Model performs well on training data but poorly on testing data.

**Q16. What is underfitting?**

**Answer:** Model fails to learn training data properly.

**Q17. What is normalization?**

**Answer:** Scaling data into smaller range.

**Q18. Why preprocessing is important?**

**Answer:** To improve model performance.

**Q19. What is training data?**

**Answer:** Data used to train the model.

**Q20. What is testing data?**

**Answer:** Data used to evaluate model performance.

## **2. Classification using Deep Neural Network**

**Q21. What is classification?**

**Answer:** Predicting categorical outputs.

**Q22. What is binary classification?**

**Answer:** Classification into two classes.

**Q23. What is multiclass classification?**

**Answer:** Classification into more than two classes.

**Q24. What is OCR?**

**Answer:** Optical Character Recognition.

**Q25. What is OCR letter recognition dataset?**

**Answer:** Dataset used for recognizing alphabets.

**Q26. What is IMDB dataset?**

**Answer:** Movie review dataset used for sentiment analysis.

**Q27. What is sentiment analysis?**

**Answer:** Finding positive or negative opinion from text.

**Q28. What is label encoding?**

**Answer:** Converting labels into numeric values.

**Q29. What is one-hot encoding?**

**Answer:** Representing categories as binary vectors.

**Q30. Which activation function is used in hidden layers?** Answer: ReLU.

**Q31. Which activation function is used in binary classification output?**  
Answer: Sigmoid.

**Q32. Which activation function is used in multiclass classification output?**  
Answer: Softmax.

**Q33. What is sigmoid function?**  
Answer: Activation producing output between 0 and 1.

**Q34. What is softmax function?**  
Answer: Converts outputs into probabilities.

**Q35. What is cross entropy loss?**  
Answer: Loss function used in classification.

**Q36. What is categorical cross entropy?**  
Answer: Loss function for multiclass classification.

**Q37. What is binary cross entropy?**  
Answer: Loss function for binary classification.

**Q38. What is accuracy?**  
Answer: Percentage of correct predictions.

**Q39. What is confusion matrix?**  
Answer: Table showing prediction results.

**Q40. What are true positives?**  
Answer: Correctly predicted positive samples.

**Q41. What are false positives?**  
Answer: Incorrectly predicted positive samples.

**Q42. What are false negatives?**  
Answer: Incorrectly predicted negative samples.

**Q43. What is precision?**  
Answer: Ratio of correct positive predictions.

**Q44. What is recall?**  
Answer: Ability to find all positive samples.

**Q45. What is F1-score?**  
Answer: Harmonic mean of precision and recall.

### 3. Convolutional Neural Network (CNN)

**Q46. What is CNN?**

**Answer:** Deep learning model mainly used for image processing.

**Q47. What is convolution?**

**Answer:** Operation used to extract image features.

**Q48. What is kernel/filter?**

**Answer:** Small matrix used for feature extraction.

**Q49. What is feature map?**

**Answer:** Output produced after convolution.

**Q50. What is pooling?**

**Answer:** Reducing image dimensions.

**Q51. What is max pooling?**

**Answer:** Selecting maximum value from region.

**Q52. What is flattening?**

**Answer:** Converting matrix into vector.

**Q53. What is dropout?**

**Answer:** Technique used to reduce overfitting.

**Q54. What is image classification?**

**Answer:** Categorizing images into classes.

**Q55. What is MNIST Fashion Dataset?**

**Answer:** Dataset containing fashion clothing images.

**Q56. What is plant disease detection?**

**Answer:** Identifying plant diseases using images.

**Q57. Why CNN is suitable for images?**

**Answer:** Automatically extracts spatial features.

**Q58. What is image preprocessing?**

**Answer:** Resizing and normalizing images.

**Q59. What is data augmentation?**

**Answer:** Generating modified training images.

**Q60. What are epochs in CNN?**

**Answer:** Iterations during training.

**Q61. What is stride?**

**Answer:** Step size during convolution.

**Q62. What is padding?**

**Answer:** Adding extra pixels around image borders.

**Q63. What is grayscale image?**

**Answer:** Image having shades of gray only.

**Q64. What is RGB image?**

**Answer:** Image containing red, green, and blue channels.

**Q65. Applications of CNN?**

**Answer:** Medical imaging, face recognition, autonomous vehicles.

## **4. Recurrent Neural Network (RNN) – Google Stock Price Prediction**

**Q66. What is RNN?**

**Answer:** Neural network designed for sequential data.

**Q67. What is time series data?**

**Answer:** Data collected over time intervals.

**Q68. Why RNN is used for stock prediction?**

**Answer:** RNN remembers previous information.

**Q69. What is sequence data?**

**Answer:** Ordered data points.

**Q70. What is hidden state in RNN?**

**Answer:** Memory of previous inputs.

**Q71. What is Google stock dataset?**

**Answer:** Historical stock prices of Google.

**Q72. What is stock price prediction?**

**Answer:** Forecasting future stock prices.

**Q73. What is LSTM?**

**Answer:** Long Short-Term Memory network.

**Q74. Why LSTM is better than basic RNN?**

**Answer:** It solves vanishing gradient problem.

**Q75. What is vanishing gradient problem?**

**Answer:** Gradients become too small during training.

**Q76. What is sequential dependency?**

**Answer:** Current output depends on previous data.

**Q77. What is forecasting?**

**Answer:** Predicting future values.

**Q78. What is timestep?**

**Answer:** Single unit in sequence.

**Q79. What is sliding window?**

**Answer:** Using previous values to predict next value.

**Q80. What optimizer is commonly used in RNN?**

**Answer:** Adam optimizer.

**Q81. What loss function is used in stock**

**prediction? Answer:** Mean Squared Error.

**Q82. What is normalization in stock**

**prediction? Answer:** Scaling stock values.

**Q83. What is training sequence?**

**Answer:** Input sequence used for learning.

**Q84. Applications of RNN?**

**Answer:** Speech recognition, NLP, stock prediction.

**Q85. Difference between CNN and RNN?**

**Answer:** CNN processes images while RNN processes sequences.

## **Advanced Viva Questions**

**Q86. What is deep learning?**

**Answer:** Subset of machine learning using neural networks.

**Q87. What is machine learning?**

**Answer:** Systems learning patterns from data.

**Q88. Difference between AI, ML and DL?**

**Answer:** DL is subset of ML and ML is subset of AI.

**Q89. What is TensorFlow?**

**Answer:** Open-source deep learning framework by Google.

**Q90. What is Keras?**

**Answer:** High-level API for deep learning.

**Q91. What is PyTorch?**

**Answer:** Deep learning framework developed by Meta.

**Q92. What is GPU?**

**Answer:** Processor used for parallel computations.

**Q93. Why GPU is used in deep learning?**

**Answer:** Faster matrix calculations.

**Q94. What is backpropagation?**

**Answer:** Technique for updating neural network weights.

**Q95. What are weights in neural networks?**

**Answer:** Parameters adjusted during training.

**Q96. What are biases?**

**Answer:** Additional parameters improving learning.

**Q97. What is gradient descent?**

**Answer:** Optimization algorithm minimizing loss.

**Q98. What is learning rate?**

**Answer:** Controls weight update size.

**Q99. What is hyperparameter?**

**Answer:** Parameter set before training.

**Q100. What is model evaluation?**

**Answer:** Checking model performance.

**Q101. What is train-test split?**

**Answer:** Dividing dataset for training and testing.

**Q102. What is validation data?**

**Answer:** Data used during training evaluation.

**Q103. What is transfer learning?**

**Answer:** Using pre-trained models.

**Q104. What is early stopping?**

**Answer:** Stopping training to avoid overfitting.

**Q105. Applications of Deep Learning?**

**Answer:** Healthcare, NLP, finance, robotics, autonomous vehicles.

**Total Questions Covered: 105**